DISKO 4

⚠️ **Security Warning**: Work in an isolated environment (virtual machine or container). Never open unknown files on your primary machine.

📋 Challenge Description

Can you find the flag in this disk image? This time I deleted the file! Let's see if you can get it now!

💡 Hints

- How would you look for deleted files?

🧰 Tools Used

- `7z` – Extract compressed files
- `file` – Determine file type
- `strings` – Extract printable character sequences
- `exiftool` – Read metadata
- `fdisk` – View partition tables
- `mount` – Mount a disk image as a loopback device
- `ls` – List directory contents
- `testdisk` – Recover deleted files from disk images

🔍 Complete Solution

Step 1: Download and Extract

Download the disk image named `disko-4.dd.gz` from the challenge page.

Extract the compressed disk image:

```
7z x disko-4.dd.gz
```

This decompresses the gzipped file and creates the raw disk image `disko-4.dd`.

Step 2: Basic File Inspection

Verify the file type and understand its structure:

```
file disko-4.dd
```

Expected output:

```
disko-4.dd: DOS/MBR boot sector, code offset 0x58+2, OEM-ID "mkfs.fat", Media
descriptor 0xf8, sectors/track 32, heads 8, sectors 204800 (volumes > 32 MB), FAT (32
bit), sectors/FAT 1576, reserved 0x1, serial number 0x49838d0b, unlabeled
```

Step 3: Search for Visible Text

Disk images often contain strings embedded directly in the data. Extract all readable character sequences and search for the flag:

```
strings disko-4.dd | grep -i "picoCTF"
```

Output (excerpt):

```
MESSAGE=[system] Activating via systemd: service name='org.bluez' unit='dbus-
org.bluez.service' requested by ':1.81' (uid=1000 pid=12105 comm="/usr/share/code/code
Desktop/picoctf-2025")
MESSAGE=[system] Activating via systemd: service name='org.bluez' unit='dbus-
org.bluez.service' requested by ':1.66' (uid=1000 pid=43129 comm="/usr/share/code/code
Desktop/picoctf-2025")
MESSAGE=[system] Activating via systemd: service name='org.bluez' unit='dbus-
org.bluez.service' requested by ':1.65' (uid=1000 pid=2141 comm="/usr/share/code/code
Desktop/picoctf-2025")
MESSAGE=[system] Activating via systemd: service name='org.bluez' unit='dbus-
```

```
org.bluez.service' requested by ':1.65' (uid=1000 pid=2584 comm="/usr/share/code/code
Desktop/picoctf-2025")
```

No flag appears yet. We need to dig deeper.

Step 4: Inspect Metadata

Examine the disk image metadata:

```
exiftool disko-4.dd
```

Expected output:

```
ExifTool Version Number         : 13.50
File Name                       : disko-4.dd
Directory                       : .
File Size                       : 105 MB
File Modification Date/Time     : 2026:05:09 15:09:36+02:00
File Access Date/Time           : 2026:05:09 15:09:36+02:00
File Inode Change Date/Time     : 2026:05:09 15:09:36+02:00
File Permissions                : -rw-r--r--
Error                           : Unknown file type
```

Nothing suspicious here.

Step 5: View Partitions

Use `fdisk` to see the partition structure:

```
fdisk -l disko-4.dd
```

Expected output:

```
Disk disko-4.dd: 100 MiB, 104857600 bytes, 204800 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
```

```
Disklabel type: dos
Disk identifier: 0x00000000
```

There are **no explicit partitions**, which hints that a filesystem is written directly to the image (a “superfloppy” or raw filesystem image).

Step 6: Mount the Disk Image

Since the image contains a FAT32 filesystem directly (no partition table), we can mount it as a loopback device:

```
sudo mkdir /mnt/disko-4


sudo mount -o loop disko-4.dd /mnt/disko-4
```

> **Note**: The `-o loop` option allows Linux to treat a regular file as a block device.

Step 7: Explore the Mounted Image

After mounting, list its contents:

```
ls /mnt/disko-4
```

Output:

```
log
```

Let’s check inside the `log/` directory:

```
ls -l /mnt/disko-4/log
```

Output:

```
total 1607
-rwxr-xr-x 1 root root  21322 Mar 31  2025 Xorg.0.log
-rwxr-xr-x 1 root root  20753 Mar 31  2025 Xorg.0.log.old
-rwxr-xr-x 1 root root      0 Mar 31  2025 alternatives.log
```

```
-rwxr-xr-x 1 root root    194 Mar 31  2025 alternatives.log.2.gz
drwxr-xr-x 2 root root   1536 Mar 31  2025 apt
-rwxr-xr-x 1 root root      0 Mar 31  2025 boot.log
-rwxr-xr-x 1 root root   7563 Mar 31  2025 boot.log.1
-rwxr-xr-x 1 root root   7125 Mar 31  2025 boot.log.5
-rwxr-xr-x 1 root root   7420 Mar 31  2025 boot.log.6
-rwxr-xr-x 1 root root 299509 Mar 31  2025 daemon.log
-rwxr-xr-x 1 root root  24374 Mar 31  2025 debug
-rwxr-xr-x 1 root root   3378 Mar 31  2025 dpkg.log.1
-rwxr-xr-x 1 root root      0 Mar 31  2025 dpkg.log.2.gz
-rwxr-xr-x 1 root root 161557 Mar 31  2025 dpkg.log.4.gz
-rwxr-xr-x 1 root root 259660 Mar 31  2025 dpkg.log.5.gz
-rwxr-xr-x 1 root root  32032 Mar 31  2025 faillog
drwxr-xr-x 3 root root    512 Mar 31  2025 inetsim
drwxr-xr-x 3 root root   1024 Mar 31  2025 installer
drwxr-xr-x 3 root root    512 Mar 31  2025 journal
....
```

> **Observation**: Nothing obvious appears here. However, the challenge description states that the file was **deleted**, so we need to look for deleted files.

Step 8: Recover Deleted Files with TestDisk

To recover deleted files from a disk image, we can use forensic tools like `TestDisk` (or `The Sleuth Kit`), which allow us to analyze the filesystem structure and retrieve files that are no longer accessible via standard mounting.

```
sudo testdisk disko-4.dd
```

Steps to follow in TestDisk:

```
TestDisk 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

  TestDisk is free software, and
comes with ABSOLUTELY NO WARRANTY.

Select a media and choose 'Proceed' using arrow keys:
>Disk disko-4.dd - 104 MB / 100 MiB

>[Proceed ]  [  Quit  ]

Note:
Disk capacity must be correctly detected for a successful recovery.
If a disk listed above has an incorrect size, check HD jumper settings and BIOS
detection, and install the latest OS patches and disk drivers.
```

```
TestDisk 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org


Disk disko-4.dd - 104 MB / 100 MiB

Please select the partition table type, press Enter when done.
 [Intel  ] Intel/PC partition
 [EFI GPT] EFI GPT partition map (Mac i386, some x86_64...)
 [Humax  ] Humax partition table
 [Mac    ] Apple partition map (legacy)
>[None   ] Non partitioned media
 [Sun    ] Sun Solaris partition
 [XBox   ] XBox partition
 [Return ] Return to disk selection


Hint: None partition table type has been detected.
```

```
TestDisk 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk disko-4.dd - 104 MB / 100 MiB - CHS 800 8 32

     Partition                  Start        End    Size in sectors
>   P FAT32                    0   0  1   799   7 32     204800 [NO NAME]
```

```
TestDisk 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk disko-4.dd - 104 MB / 100 MiB - CHS 800 8 32
     Partition                  Start        End    Size in sectors
   P FAT32                    0   0  1   799   7 32     204800 [NO NAME]

Boot sector
OK

Backup boot sector
OK

First sectors (boot code and partition information) are not identical.

A valid FAT Boot sector must be present in order to access
any data; even if the partition is not bootable.

 [  Quit  ] >[  List  ] [Org. BS ]  [Backup BS]  [Rebuild BS]  [  Dump   ]
          List directories and files, copy and undelete data from FAT
```

```
TestDisk 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org
   P FAT32                    0   0  1   799   7 32     204800 [NO NAME]
Directory /

>drwxr-xr-x     0     0         0 31-Mar-2025 15:32 log



                                                   Next
Use Right to change directory, 'h' to hide deleted files
    'q' to quit, ':' to select the current file, 'a' to select all files
    'C' to copy the selected files, 'c' to copy the current file
```

```
TestDisk 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org
   P FAT32                    0   0  1   799   7 32     204800 [NO NAME]
Directory /log
                                                   Previous
 -rwxr-xr-x     0     0     21322 31-Mar-2025 15:31 Xorg.0.log
 -rwxr-xr-x     0     0       195 31-Mar-2025 15:31 vmware-network.4.log
 -rwxr-xr-x     0     0         0 31-Mar-2025 15:31 boot.log
 -rwxr-xr-x     0     0     46637 31-Mar-2025 15:31 syslog.3.gz
 -rwxr-xr-x     0     0     19624 31-Mar-2025 15:31 vmware-vmtoolsd-root.log
 -rwxr-xr-x     0     0    299509 31-Mar-2025 15:31 daemon.log
 -rwxr-xr-x     0     0    704371 31-Mar-2025 15:31 messages
 -rwxr-xr-x     0     0       194 31-Mar-2025 15:31 alternatives.log.2.gz
 -rwxr-xr-x     0     0     24374 31-Mar-2025 15:31 debug
 drwxr-xr-x     0     0         0 31-Mar-2025 15:31 lightdm
 -rwxr-xr-x     0     0       193 31-Mar-2025 15:31 vmware-network.6.log
 -rwxr-xr-x     0     0         0 31-Mar-2025 15:31 alternatives.log
 -rwxr-xr-x     0     0      3007 31-Mar-2025 15:31 vmware-vmsvc-root.1.log
 -rwxr-xr-x     0     0     20753 31-Mar-2025 15:31 Xorg.0.log.old
 -rwxr-xr-x     0     0       195 31-Mar-2025 15:31 vmware-network.8.log
 -rwxr-xr-x     0     0      2292 31-Mar-2025 15:31 vmware-vmsvc-root.log
 -rwxr-xr-x     0     0      3304 31-Mar-2025 15:31 vmware-vmsvc-root.3.log
 -rwxr-xr-x     0     0      7420 31-Mar-2025 15:31 boot.log.6
 -rwxr-xr-x     0     0       193 31-Mar-2025 15:31 vmware-network.5.log
 -rwxr-xr-x     0     0        59 31-Mar-2025 15:31 macchanger.log.4.gz
 drwxr-xr-x     0     0         0 31-Mar-2025 15:31 apt
 -rwxr-xr-x     0     0      3378 31-Mar-2025 15:31 dpkg.log.1
 -rwxr-xr-x     0     0      7125 31-Mar-2025 15:31 boot.log.5
 -rwxr-xr-x     0     0     87168 31-Mar-2025 15:31 wtmp
 -rwxr-xr-x     0     0    259660 31-Mar-2025 15:31 dpkg.log.5.gz
 -rwxr-xr-x     0     0    292292 31-Mar-2025 15:31 lastlog
 -rwxr-xr-x     0     0       195 31-Mar-2025 15:31 vmware-network.3.log
 -rwxr-xr-x     0     0    288914 31-Mar-2025 15:31 syslog.4.gz
 -rwxr-xr-x     0     0      7563 31-Mar-2025 15:31 boot.log.1
 -rwxr-xr-x     0     0       195 31-Mar-2025 15:31 vmware-network.2.log
 -rwxr-xr-x     0     0     32032 31-Mar-2025 15:31 faillog
 -rwxr-xr-x     0     0     18801 31-Mar-2025 15:31 kern.log.3.gz
 -rwxr-xr-x     0     0    161557 31-Mar-2025 15:31 dpkg.log.4.gz
 -rwxr-xr-x     0     0       195 31-Mar-2025 15:31 vmware-network.1.log
 -rwxr-xr-x     0     0       195 31-Mar-2025 15:31 vmware-network.7.log
 drwxr-xr-x     0     0         0 31-Mar-2025 15:31 journal
 -rwxr-xr-x     0     0         0 31-Mar-2025 15:31 vmware-network.log
 -rwxr-xr-x     0     0         0 31-Mar-2025 15:31 dpkg.log.2.gz
>-rwxr-xr-x     0     0        87  4-Feb-2026 21:54 dont-delete.gz
                                                   Next
Use Left arrow to go back, Right to change directory, 'h' to hide deleted files
    'q' to quit, ':' to select the current file, 'a' to select all files
    'C' to copy the selected files, 'c' to copy the current file
```

> **Note**: The screenshots above show the TestDisk interface and the recovery process.

Step 9: Extract and Read the Recovered File

After successful recovery, a file named `dont-delete.gz` appears in your current directory.

First, extract the compressed file:

```
7z x dont-delete.gz
```

Then read its contents:

```
cat dont-delete
```

Output:

```
Here is your flag
picoCTF{<REDACTED>}
```

✅ Final Result

The flag is successfully recovered from the deleted file:

```
picoCTF{<REDACTED>}
```

> **Note**: The actual flag is redacted here. In the real challenge, the exact value will be displayed.

🧠 Key Concepts

- **Raw Disk Image**: A sector-by-sector copy of a storage device, which can contain a filesystem without a partition table.
- **Loopback Mount**: Mounting a file as a block device using `mount -o loop`.
- **FAT32 Filesystem**: A simple, widely supported filesystem often used on removable media.
- **File Deletion**: When a file is deleted, its data often remains on disk until overwritten. Only the filesystem entry (metadata) is removed.
- **TestDisk**: A powerful open-source tool for recovering lost partitions and undelete files.
- **Forensic Recovery**: Using specialized tools to recover data that is no longer visible through standard filesystem access.